

# PharmaSync V4: Unified Clinical Network

## Technical Specification & Architecture

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## 1 Introduction

PharmaSync V4 is an advanced, geolocation-aware pharmacy management ecosystem. Unlike traditional inventory systems, V4 implements a dynamic **Proposal-based Economy** (Uber-like model) where patients, pharmacists, and suppliers negotiate prices and quantities in real-time.

## 2 System Architecture

The system is built on a Node.js Express backend with a MySQL relational database. The frontend is a sophisticated Single Page Application (SPA) utilizing CSS Grid and modern JavaScript.

## 3 Economic Model: The Proposal System

Fixed price lists have been deprecated in favor of a **Negotiated Proposal** model.

- **Patient to Pharmacy:** Patients broadcast a medicine requirement with a target price. Pharmacies accept or reject based on stock and profitability.
- **Pharmacy to Supplier:** Pharmacists propose restock batches to suppliers. The system provides a “System Rating” (Excellent, Good, Fair, Poor) to advise on market competitiveness.

## 4 Database Schema

The relational model consists of five primary entities: Users, Patients, Pharmacies, Medicines, and Proposals. Geolocation data (latitude and longitude) is stored with 8-decimal precision to ensure high accuracy.

## 5 UI/UX Philosophy

The visual identity follows a **Clean Clinical** aesthetic. The design is optimized for high-density information display, removing non-professional elements such as emojis and utilizing the full screen width (wrap-around layout) for complex data tables and analytics.